School of Informatics & Cybersecurity - Assessment Submission Coversheet

Student Information

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Module Information

Module Title:	Network Security
Module Code:	H1001

Assessment Information

Assessment Title:	Open Source Firewalls

Declaration

☐ I confirm that I have read and understood the University's Academic Integrity Policy.

https://www.tudublin.ie/explore/about-the-university/academic-affairs/academic-quality-assurance-and-enhancement/academic-integrity/

☐ I have completed the TU Dublin AI Driving License theory test.

https://www.tudublin.ie/explore/about-the-university/academic-affairs/lta/lta-resources/artificial-intelligence-driving-license/

☐ I declare that this assessment submission is my own work.

☐ I confirm that I have read and understood the AI traffic light system used as part of this assessment brief and I have adhered to the level of AI usage allowed for this assessment.

Student signature:



Date: 27/10/2025

Introduction

The purpose of this report is to evaluate and research the effectiveness, usability and the overall security features of three open-source firewalls. The three firewalls being pfSense, OPNsense and IPFire. We will have the firewalls connected to Virtual Machines that are all setup the same to ensure the results are fair and equal. The firewalls will be compared while they act as the first line of defence under attack by another Virtual Machine. The testing done focused on how easy it was to configure each firewall as a user and navigating through the firewall's features, the rule management and the ability to stop penetration from an attacker, the chosen pen test being brute-force.

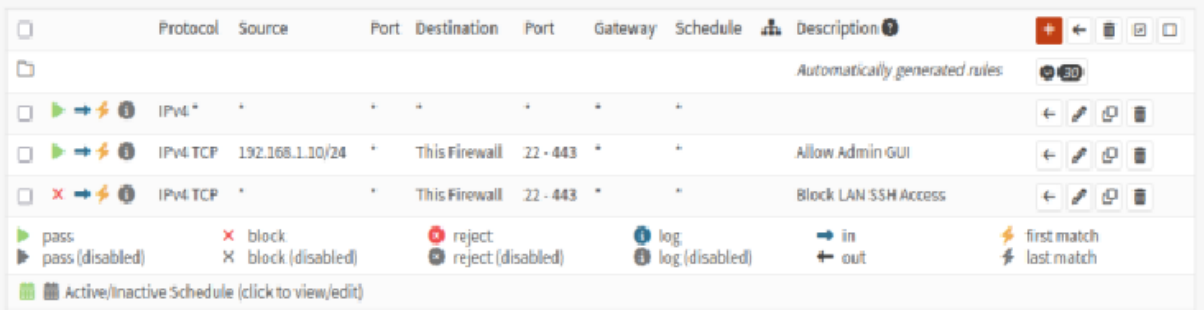
This report will include the setup process for each firewall, the configuration, and then the results of the penetration. The results will highlight the strengths, weaknesses and usability differences between the three firewalls.

Tevin's Part

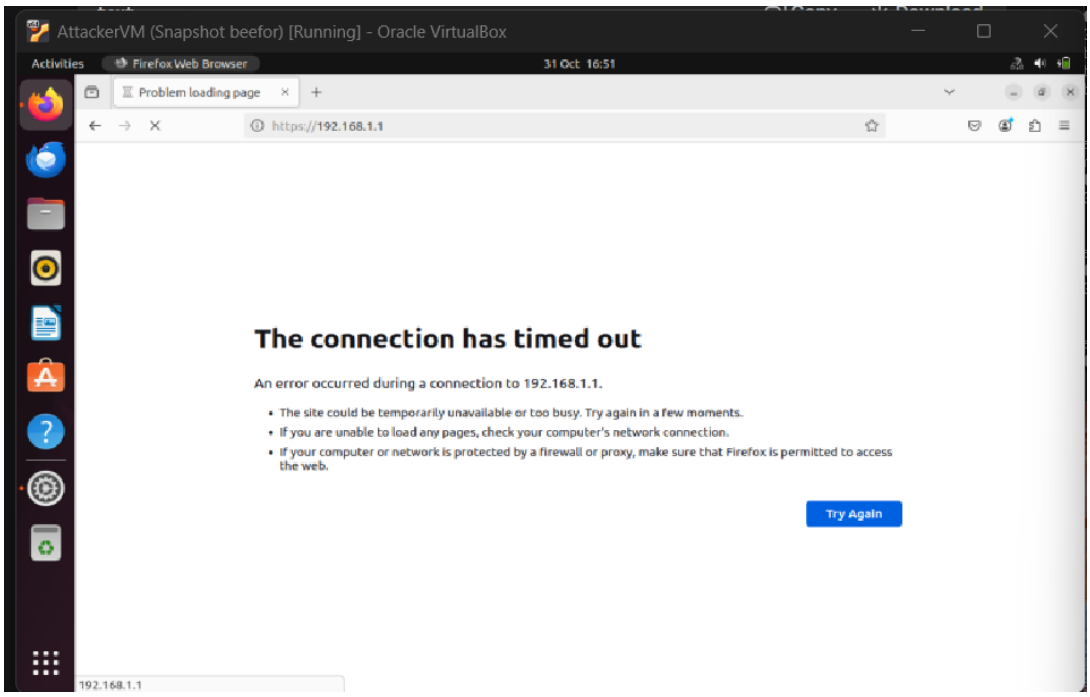
My firewall of choice was OPNsense. OPNsense is an open-source firewall, it is mainly used to secure and mange network traffic through features such as packet filtering, NAT, intrusion detection etc. It comes with a web-based GUI for any configuration a user may want to, for any plugins for more improved firewall capabilities. Generally, there is a bit of a learning curve with using OPNsense since you do have to know the basics of networking, but it is a lot easier to use than most other firewalls.

I wanted to see how easy it was for a user to use each Firewall to block communication from untrusted VM's. For OPNsense this was done by adding rules that permit only my "client"/administration VM's IP to reach the destination of "this firewall" in the range of the SSH port (22) – GUI port (443). Then adding a separate rule that blocks all other

hosts on the LAN network.



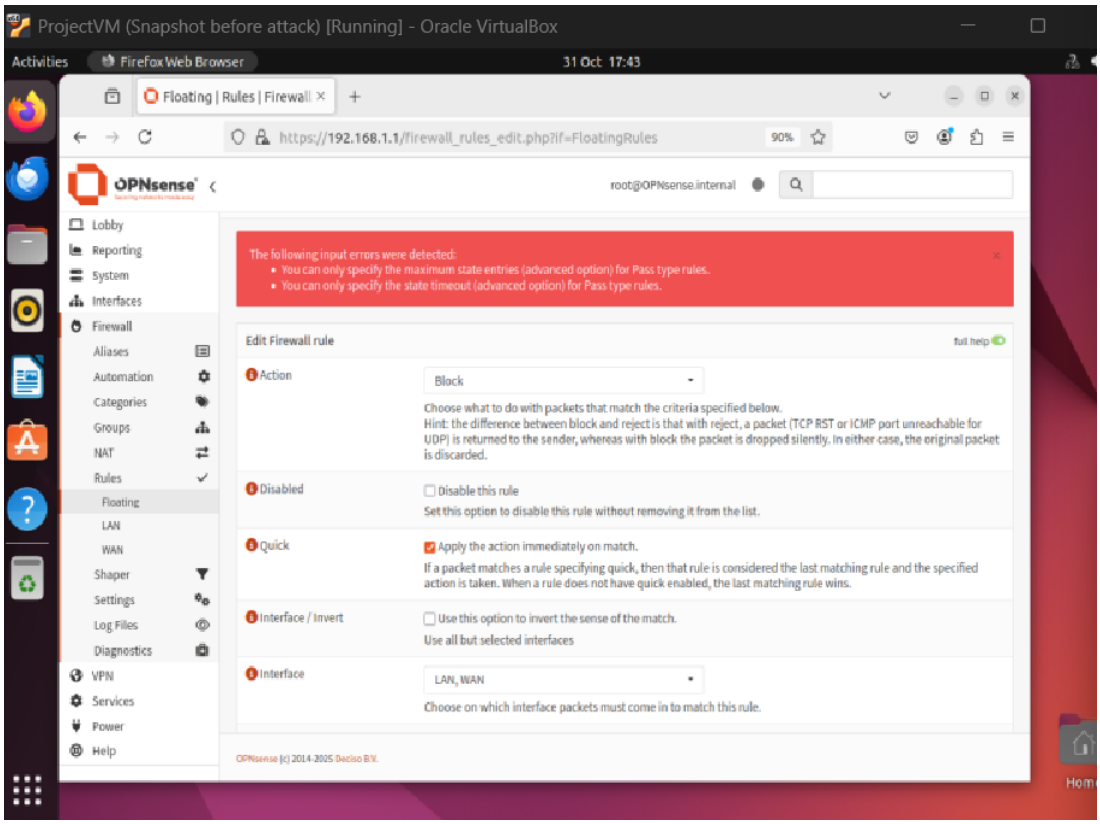
This rule simply prevents the “attacker” from accessing the GUI completely. It fully stops communication to a level where any communication attacks the attacker may try will fail 10/10 times.



Interface	Time	Source	Destination	Proto	Label
LAN	2025-10-31T16:00:33	192.168.10.0	192.168.1.1	icmp	Default deny / state violation rule
LAN	2025-10-31T16:00:32	192.168.10.0	192.168.1.1	icmp	Default deny / state violation rule
LAN	2025-10-31T16:00:31	192.168.10.0	192.168.1.1	icmp	Default deny / state violation rule
WAN	2025-10-31T16:00:30	192.168.8.4:53689	192.168.8.255:15600	udp	Block private networks from WAN
LAN	2025-10-31T16:00:30	192.168.10.0	192.168.1.1	icmp	Default deny / state violation rule
LAN	2025-10-31T16:00:29	192.168.10.0	192.168.1.1	icmp	Default deny / state violation rule
LAN	2025-10-31T16:00:29	192.168.10.0	192.168.1.1	icmp	Default deny / state violation rule
LAN	2025-10-31T16:00:27	192.168.10.0	192.168.1.1	icmp	Default deny / state violation rule
LAN	2025-10-31T16:00:26	192.168.10.0	192.168.1.1	icmp	Default deny / state violation rule
LAN	2025-10-31T16:00:25	192.168.10.0	192.168.1.1	icmp	Default deny / state violation rule
LAN	2025-10-31T16:00:24	192.168.10.0	192.168.1.1	icmp	Default deny / state violation rule
WAN	2025-10-31T16:00:24	192.168.8.4:47451	192.168.8.255:15600	udp	Block private networks from WAN
LAN	2025-10-31T16:00:23	192.168.10.0	192.168.1.1	icmp	Default deny / state violation rule
LAN	2025-10-31T16:00:23	[fe80::3e8c:46e6:b47c:1d58]:546	[ff02::1:2]:547	udp	allow access to DHCPv6 server on LAN
LAN	2025-10-31T16:00:22	192.168.10.0	192.168.1.1	icmp	Default deny / state violation rule
LAN	2025-10-31T16:00:21	192.168.10.0	192.168.1.1	icmp	Default deny / state violation rule
LAN	2025-10-31T16:00:21	192.168.1.10:57702	192.168.1.1:443	tcp	anti-lockout rule

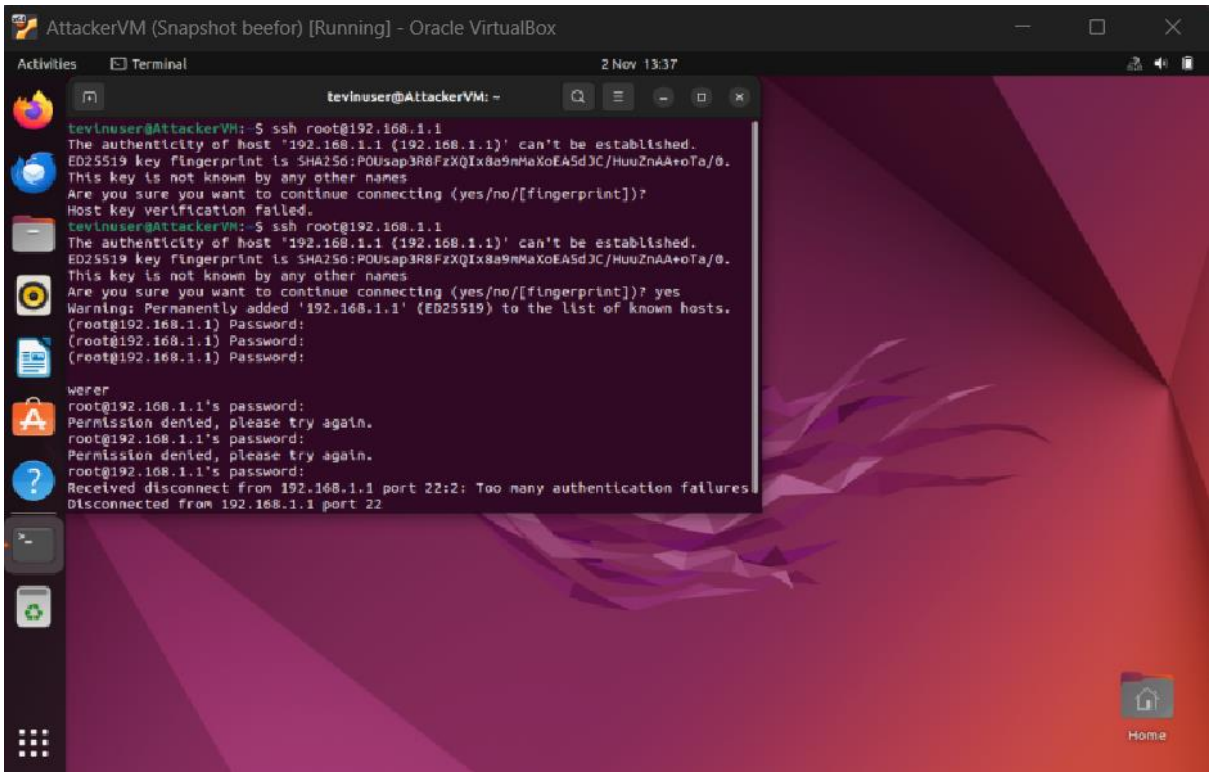
This log shows all attempts from the “attackers” attempts to try and communicate with the host, and as it is clearly labelled, it was denying access since it violates the rules.

I tried a brute force attack to see how well OPNsense will deal with stopping the attacker from accessing the GUI if they tried. No matter how far into the community notes or even community pages such as reddit I went through I couldn’t at all see a rule set to limit the number of times an IP can try the password. The more research I did, the more I saw how tedious it was to try and add a simple password try limit to OPNsense.



The rules don't allow you to add a "maximum state" and a "maximum stale timeout" to any Block type rules. The maximum state rule would've made it so the person trying to login only has a set amount of tries to type up the password, let's say 5 simultaneous attempts, and the maximum state timeout means that OPNsense would've remembered each attempt for a set amount of time (60 second). This was due to the way OPNsense recognises each rule type. After a bit of research, I saw that in pass type rules, OPNsense acts as a security guard who lets people in and remembers their IP, while the block type rule simply doesn't care for a "persons" IP and simply refuses access to wherever they were trying to enter. This saves a ton of time and resources due to how simple and concise the block rule is (OPNsense Document 2025). What I found a lot of people within the OPNsense community tend to do is simply not exposing their GUI to the internet and it is only accessible through a VPN (OPNsense Forum). VPNs use certificate and passwords causing them to be much harder to try and brute force through, but, due to my limited resources using VM's I couldn't try this out and see how strong of a brute force preventor this setup was.

The only trouble I had with trying to set a password limit was within the OPNsense SSH for the GUI, but, I found that if I tried to login to the OPNsense CLI through terminal, using the "ssh root@192.168.1.1" command it allowed me to try and access the login for the CLI. I then tried a "faux" brute force test, this led to my attacker VM to be disconnected due to "too many authentication failures after around 6 authentication attempts.



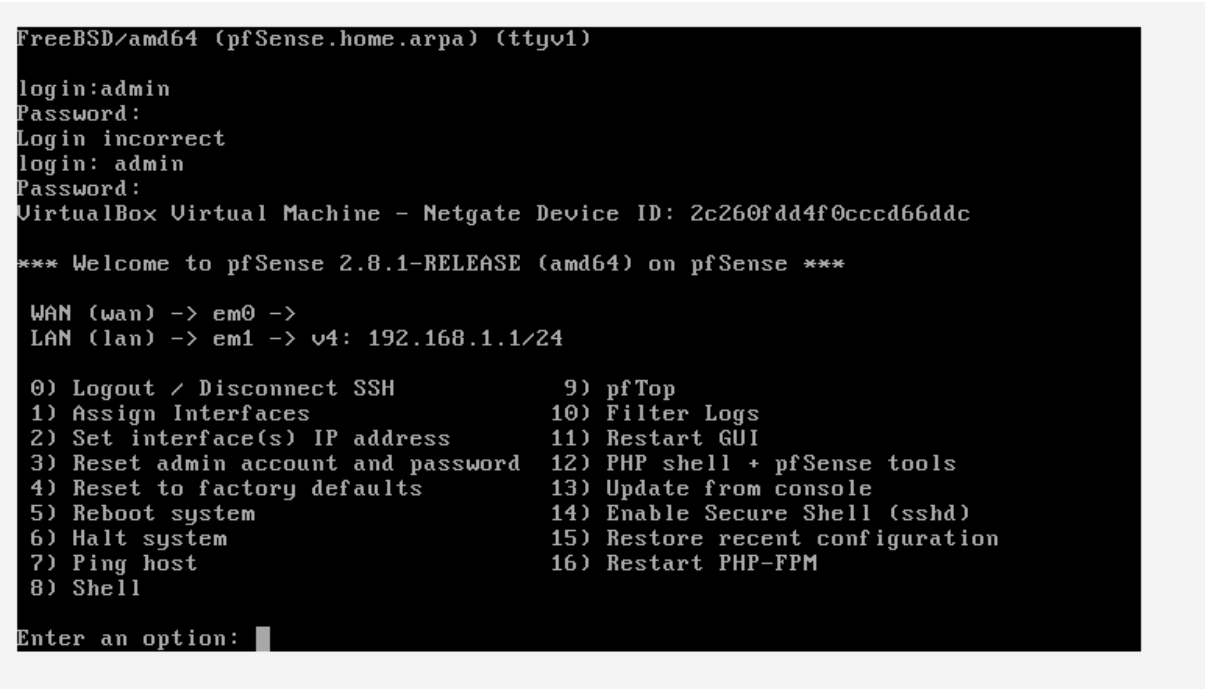
Overall, OPNsense is a great firewall with all things considered, I would say it preformed fairly well as a gateway firewall, as it allowed me to apply simple yet effective rules, such as allowing only the set admin IP to be able to access the GUI and blocking any other IP's from getting access. I used the very easy to access and navigate, packeting filter to reliably enforce access controls and generate usable logs to see the denied attempts. However, OPNsense didn't automatically block the malicious IP address, I could've manually done this using the GUI, so I will take a few points off for the tedious nature of this.

The default rule model exposes a very limited application method for the user, as there isn't a single built in GUI configuration to block an IP that has had a few failed SSH attempts as I couldn't set a limit to the "max tries" or even set a lockout. This is because any block rule outright stops access to whatever destination is set on it, but it doesn't allow you to set a stateful floating rule to lock them out for a period after a set amount of tries. I found that this was only for the GUI, but when attempting to brute force into the CLI's SSH the OPNsense firewall will disconnect the attacker and send them out of its server.

Ken's part (B00169852)

In this part of my research on firewalls, I will be choosing a firewall called pfSense. It is an open-source firewall system that runs on a dedicated computer, turning it into a powerful network security device. So, in this, I will be setting it up and telling you about my experience with it.

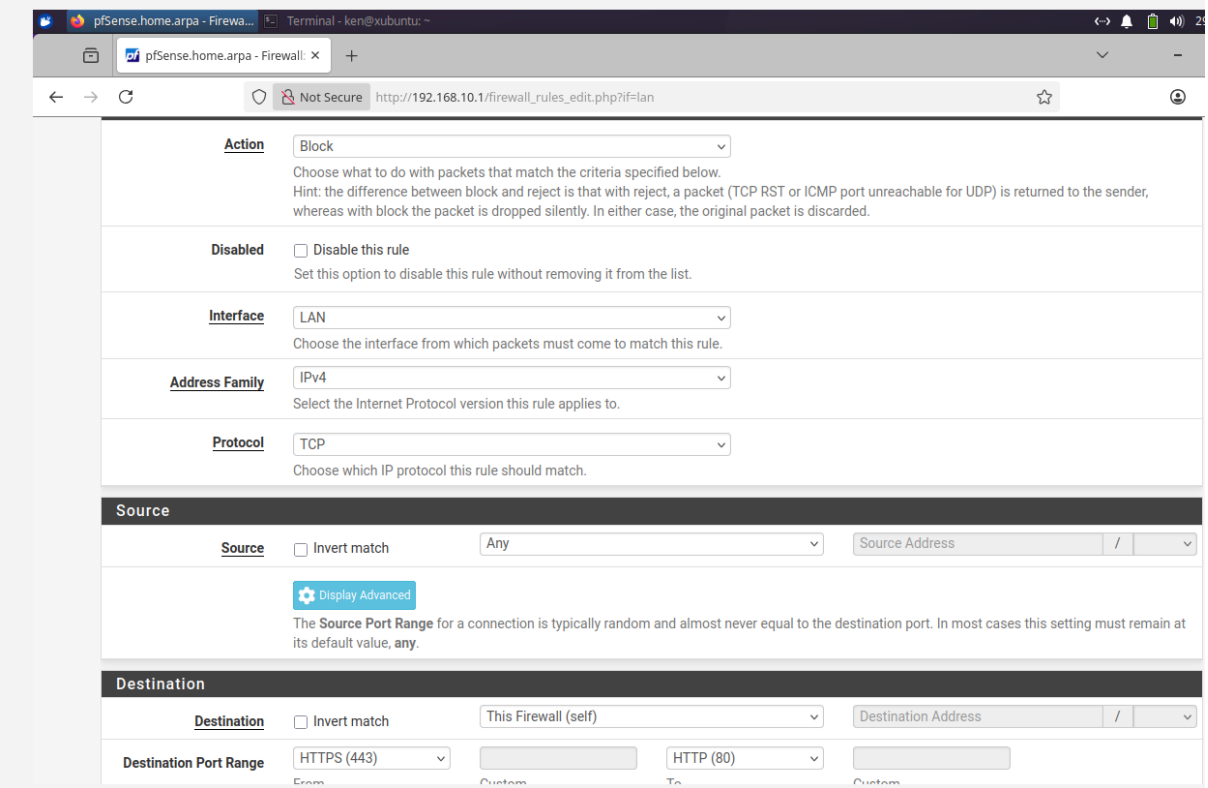
When I started the software, it was a simple setup. I set up the WAN and LAN, but I had a lot of issues. On the LAN, it would just refuse to connect to the internet.



I had to access the GUI, and the rules had to allow the LAN traffic. This was relatively easy, as the GUI was easy to configure.

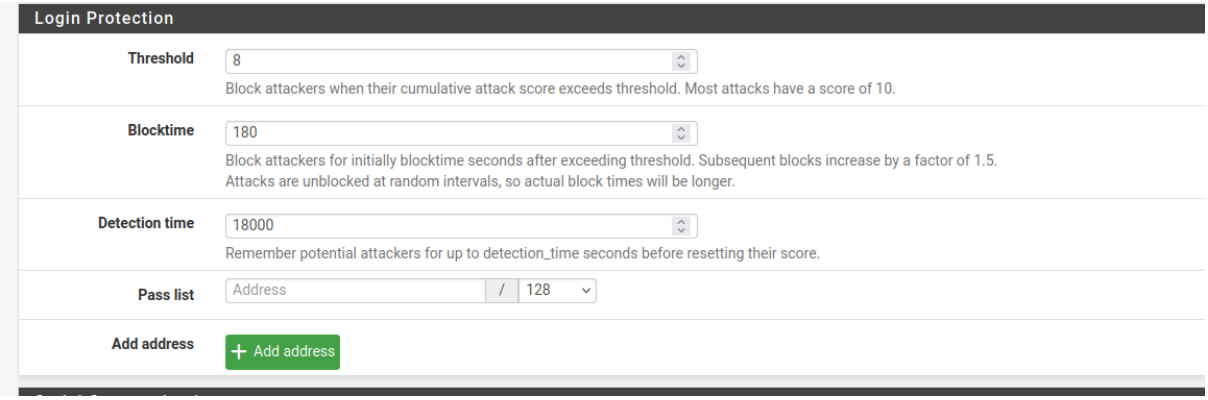
However, there is still one issue: any PC on the network can access the GUI. So, I had to change it so that only the admin PC could access it. I added a rule that prevents anyone else from using HTTP and HTTPS to reach the firewall. This will prevent people from

using the GUI unless they have the correct IP.



The attacker PC can ping the firewall and realize it can SSH into the system if the password is known. So, I had to make another rule where port 22 is only accessible to the admin account. This was easy to set up thanks to the GUI.

The good thing about pfSense is the logging and monitoring of traffic so that when there is a login successful or not it would tell me live in the machine's display. It is important that we fcan monitor logs as if there were network issues or a security breach. We would be able to find out what happened before the issues or prevent an attacker from causing more harm.

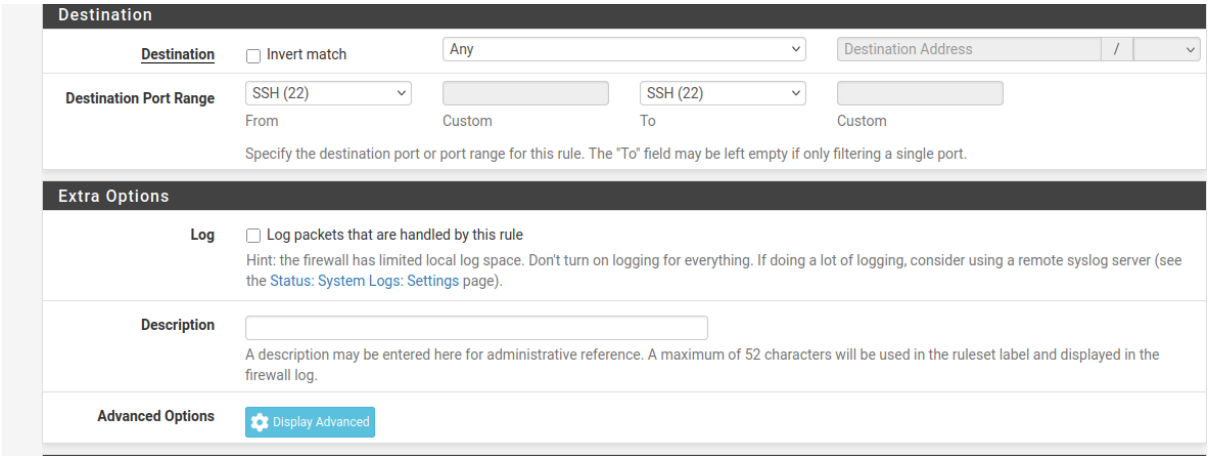


```
Enter an option:
Message from syslogd@pfSense at Oct 30 15:23:25 ...
php-fpm[423]: /index.php: Successful login for user 'admin' from: 192.168.10.10
(Local Database)

Message from syslogd@pfSense at Oct 30 17:10:54 ...
php-fpm[424]: /index.php: webConfigurator authentication error for user 'jl' fro
m: 192.168.10.10
```

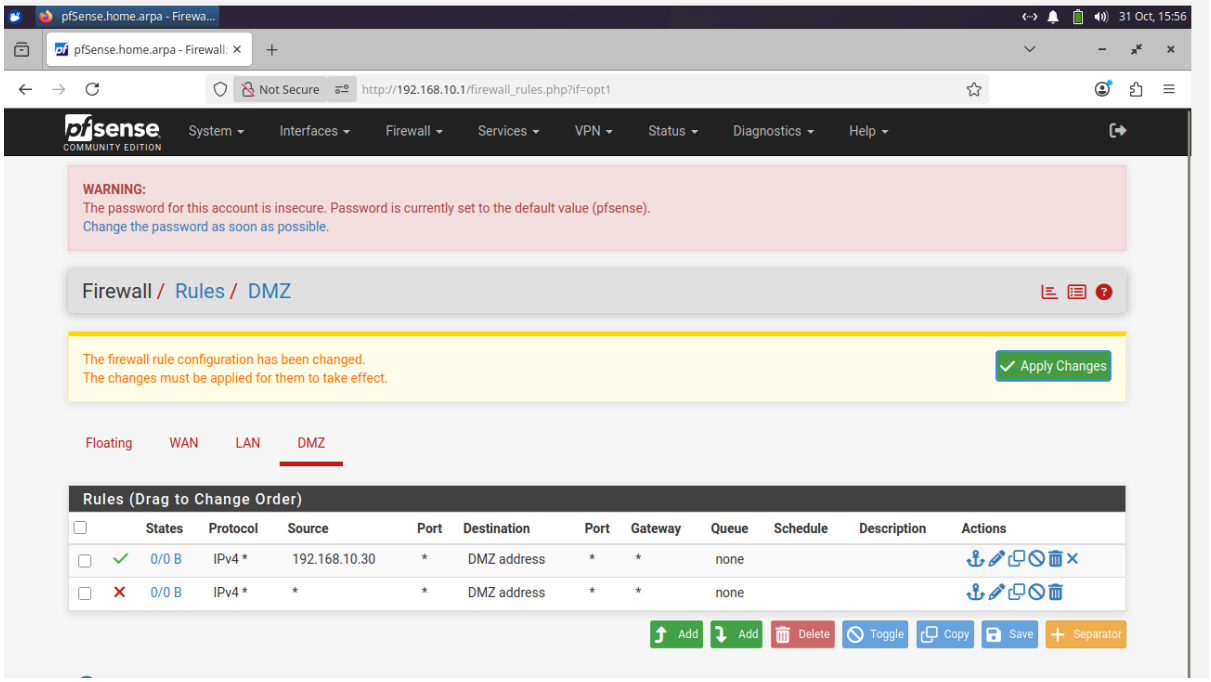
This did take a while to do, but I don't think it was the firewall itself just configuring the VMs to work with it, which was annoying. I had to go on the shell of the PC to reset and disable the firewall. A good feature of pfSense is that you can go on the terminal of the firewall and configure it directly.

I wanted more out of this firewall, so I added precautions in case someone spoofed my IP and tried to brute force their way in. You can change the settings so that if someone enters a wrong password or username a certain number of times, it will delay them and report it to the firewall. When the limit is reached, it blocks the attacker, and I could only change that setting on the firewall machine itself. Now, my attacker PC can't SSH, access the GUI, or brute force attack it.




```
ken@xubuntu:~$ ssh -p 22 admin@192.168.10.1
admin@192.168.10.1: Permission denied (publickey).
ken@xubuntu:~$
```

I decided to set up a Samba file server on a different network but attached to the firewall. Setting it so everyone can access it was very easy thanks to the GUI. All I had to do was go to my LAN port where my server is and make a rule allowing everyone to access the network. This was an important step, as it made the network feel like a proper setup. I added another VM called ENG and was able to use the files on the server. This is a security issue, as anyone whether malicious or even the admin can access and alter files.



This is where a DMZ can go in place. Setting up a DMZ was very easy ,I just isolated the port that nothing can go in or out of, only the ip of the eng pc so they can access their files. I only allowed tcp, udp, and ssh ports of the engineer's pc so he can access the file server and that's it.

Overall, me researching pf sense was probably one of the most important things I have done in my IT journey. This has taught me how to maximise and use VirtualBox. Like setting up the virtual machines and connecting them to the one network while only giving the pfSense internet access. Using the firewall itself taught me a lot on the importance of firewall as I am the one who has to configure it. During this I learned how to configure wan and lan on the Firewall, how to make them connect and talk to each other and then securing it. Securing the firewall/my network was simple but at the same time hard as this was my first time. The GUI is very easy to look at so that's a bonus. Learning to block out people from accessing the GUI and still enabling the pc to talk to each other. Configuring ssh was cool. I'm going to be honest, remotely accessing the firewall was a great experience as this is a keystone in IT. Remotely accessing and configuring networks is most of network admins jobs. Managing SSH permissions was tricky, but it taught me how to secure ssh like changing the port of ssh only allows a certain ip with verification. Setting the file server was probably the hardest thing going. Just setting it up while connecting the server to the network while everyone can access it was very tedious. Creating a DMZ for the server to be in was easy as it was just a few

clicks then add rules. I feel like other firewalls would be harder or a lot to set up or the gui would not be as good. pfSense is fully customisable and very easy to manage. More importantly is free and open source so you can get this on a network for free.

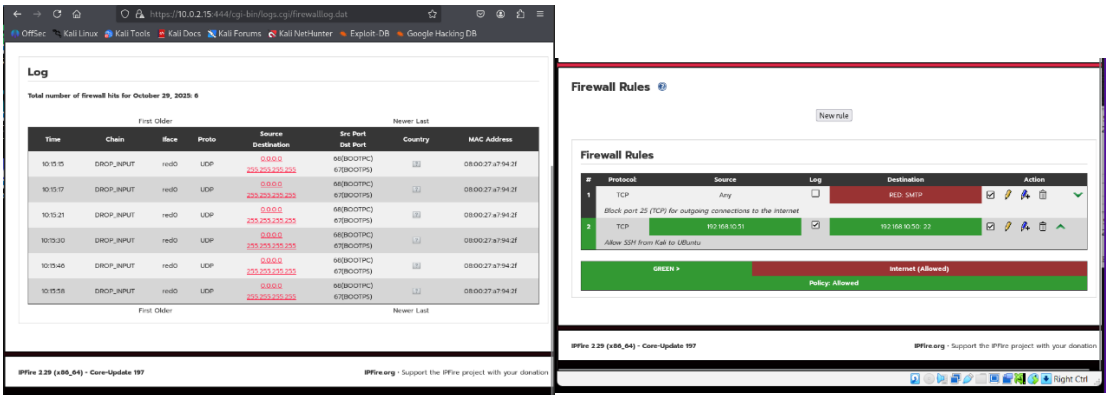
Last,and in this case least user friendly of all, we used IPFire, which introduced us to countless hurdles and complications. Built to protect, filter and manage traffic, IPFire is an open source Linux based firewall that runs on x86 hardware. Ipfire offers a web based GUI and a CLI, both have the ability to configure protections which can come in handy as it did for us.

Some key features include:

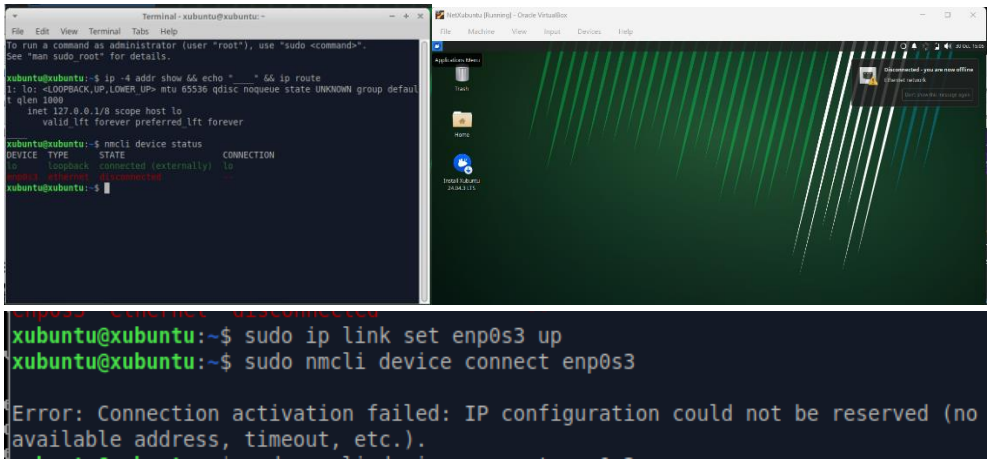
- Built-In Stateful Packet Inspection for firewalling
- Intrusion Detection/Prevention via Snort or Suricata.
- Extensible Add-On System through Pakfire.
- Dynamic DNS and multi-WAN support.
  (LinuxMind, 2025)

We aimed to configure and understand how IPFire worked and its role in a network, compared to the other Firewalls used. We were particularly interested in the ease of use for users, test its filtering capabilities & ability to block attackers from certain manouvers. Intensions were set to resolve configuration difficulties, understand what interfaces and adapters would be needed, ensure connectivity was present between the attacker, the victim and firewall systems and how we could limit them if needed. Along with these aims we sought out to understand the ease of use between the CLI and web GUI. Lastly concluding on the strengths and weaknesses of IPFire as a whole

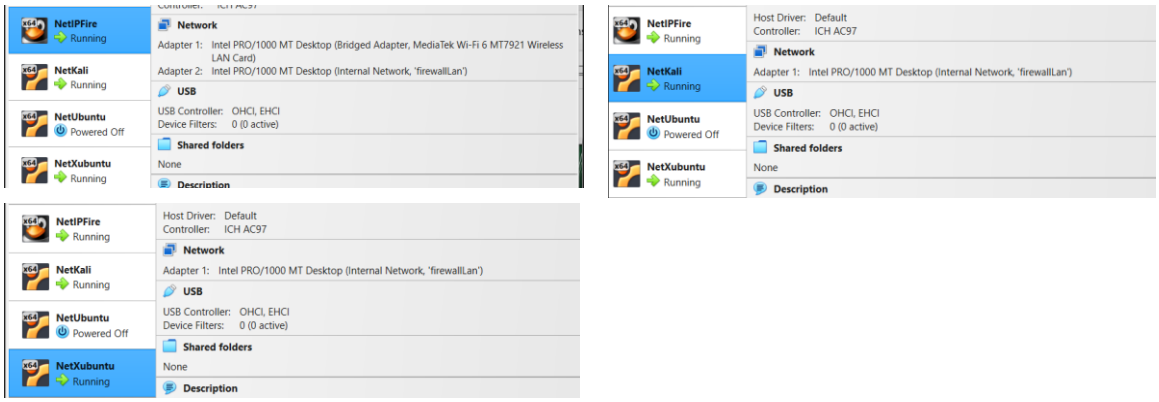
To begin, we wanted to diversify things slightly, so we used Kali Linux as the attacker, Ubuntu Server as the victim and IPFire. We originally tested a different network adapter layout also- using a NAT Network adapter for the IPFire machine along with a ‘green’ Internal Network that was also used by the two other machines.



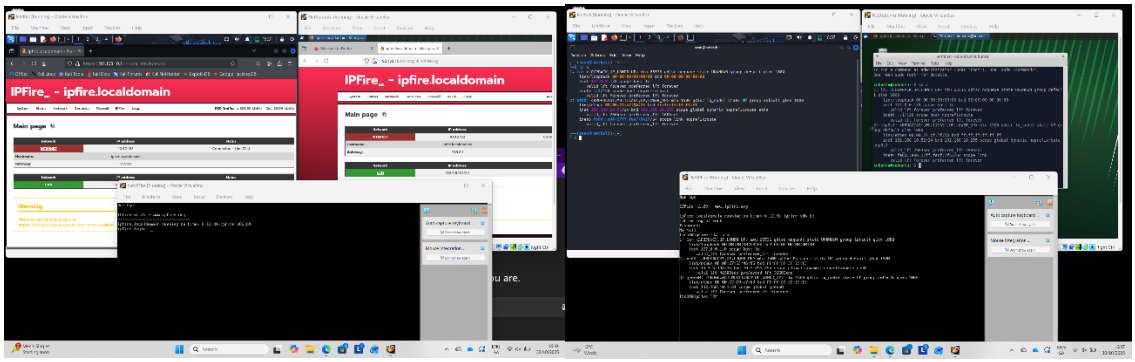
IPFire wasn't capturing anything, after setting a rule even after playing around with rules on the GUI to block ports, allow SSH and turning them on and off. Our initlal troubleshooting idea was revert back to the adapter layout we used on the previous firewalls. Just before we initiated this step we realized we were using the GUI on the attacker machine that we would have to block access to later, so we had to restart completley and replace the Ubuntu Server with Xubuntu- a costly mistake timewise.



Once setup with Xubuntu we couldn't connect to the network, so we reverted our adapter layout:



We reset IPFire settings and rebooted, configured again and we were in.

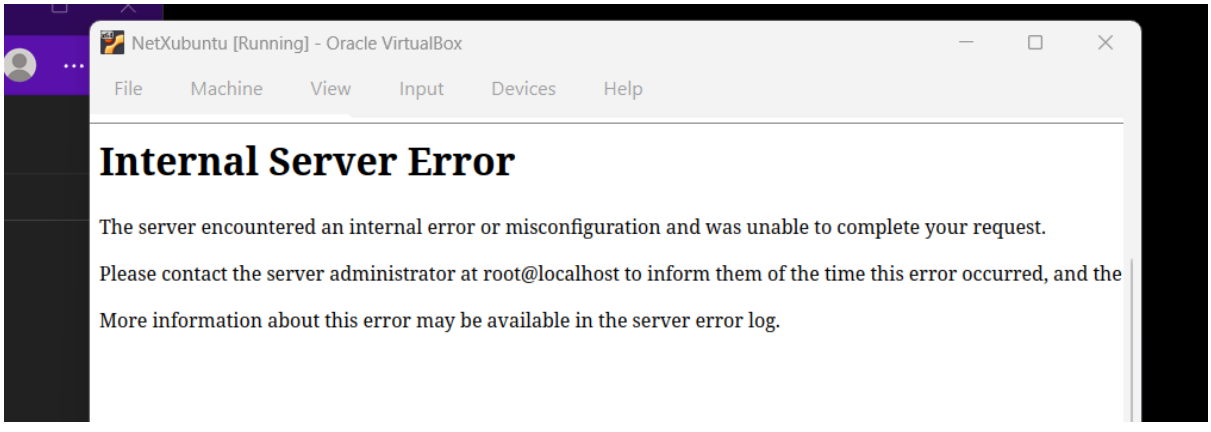



```
xubuntu@xubuntu:~$ ip a | grep inet
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host noprefixroute
    inet 192.168.10.52/24 brd 192.168.10.255 scope global dynamic noprefixroute
enp0s3
    inet6 fe80::a00:27ff:fecf:7f53/64 scope link
xubuntu@xubuntu:~$
```

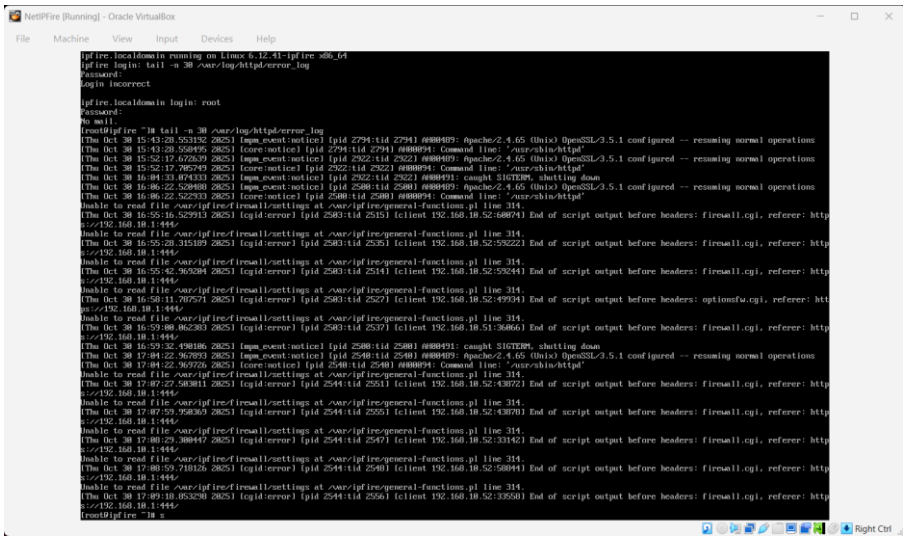
```
┌──(sean㉿netkali)-[~]
└─$ ip a | grep inet
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host noprefixroute
    inet 192.168.10.51/24 brd 192.168.10.255 scope global dynamic noprefixroute eth0
    inet6 fe80::a00:27ff:fea7:942f/64 scope link noprefixroute
```

We ensured connectivity with all machines and finally reached where we need to be to begin testing.

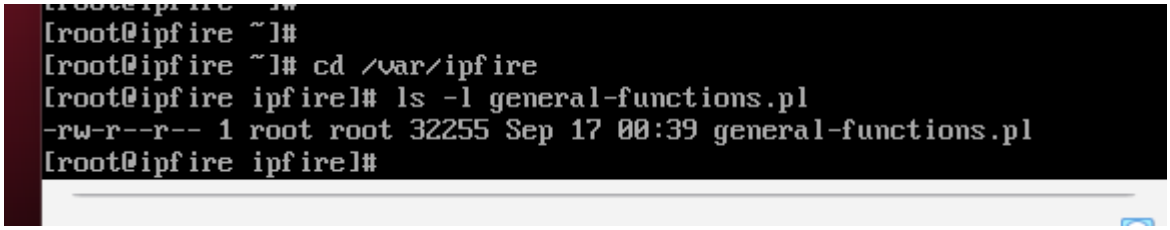
Although due to our errors, changing victim machines and troubleshhoting log capturing issues we didn't have the abundance of time we had with the other firewalls, so we narrowed our aims down to ease of use and user friendliness of IPFire.



IPFire then failed each time we should try to set firewall rules, every other part of GUI was functioning but the ability to set rules, we restarted and rebooted IPFire but nothing changed.



We checked our logs for errors and saw the general functions file wasn't being read correctly.

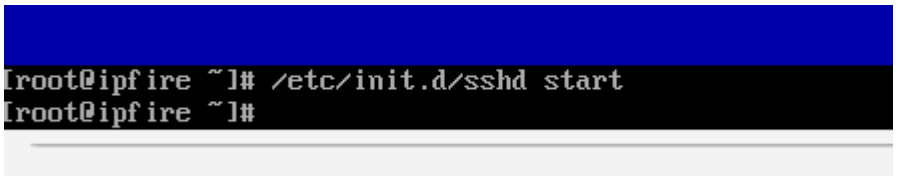


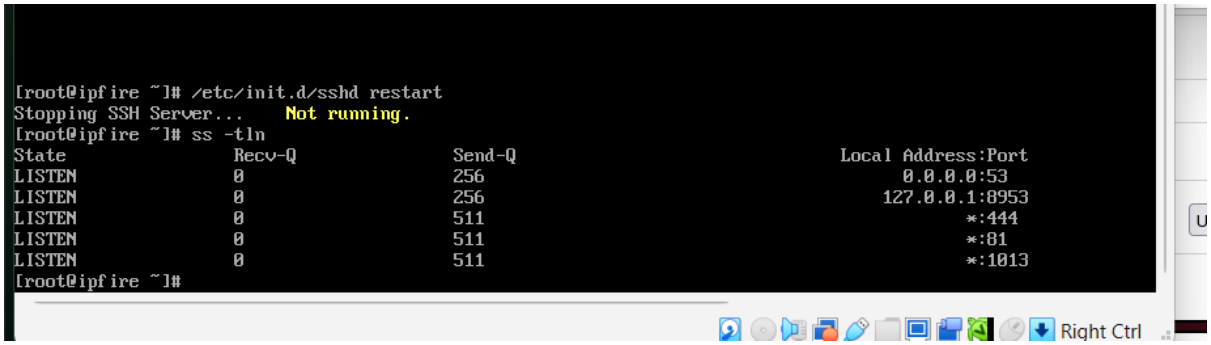
We examined the files and the 'genral functions' file was still present and had the correct permissions, so it should've been working, we surfed the web for troubleshooting techniques but nothing we came across really zeroe'd in on this issue with the setup we had. We used 'pakfire updates' command to make sure everything was updated and downloaded correctly- still no success.

(IPFire, n.d.)

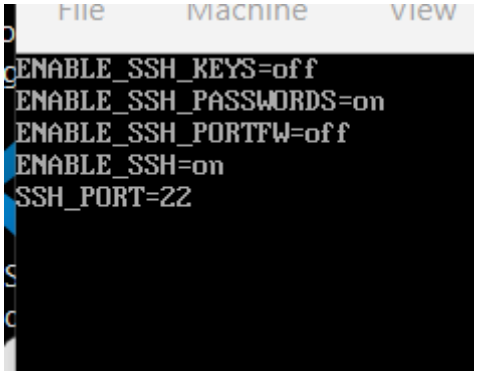
We decided to improvise and use this as an oppurtunity to diversify techniques and used the CLI of IPFire to set configurations as we were extremely tight on time at this stage due to this and previous errors.

First we decided to attempt to block the attacker from accessing the GUI as this would defeat the purpose of the Firewall.

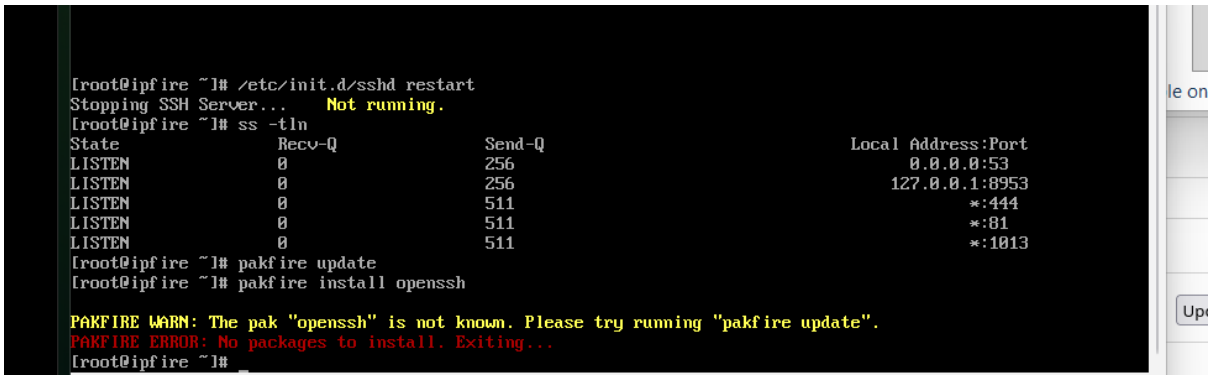




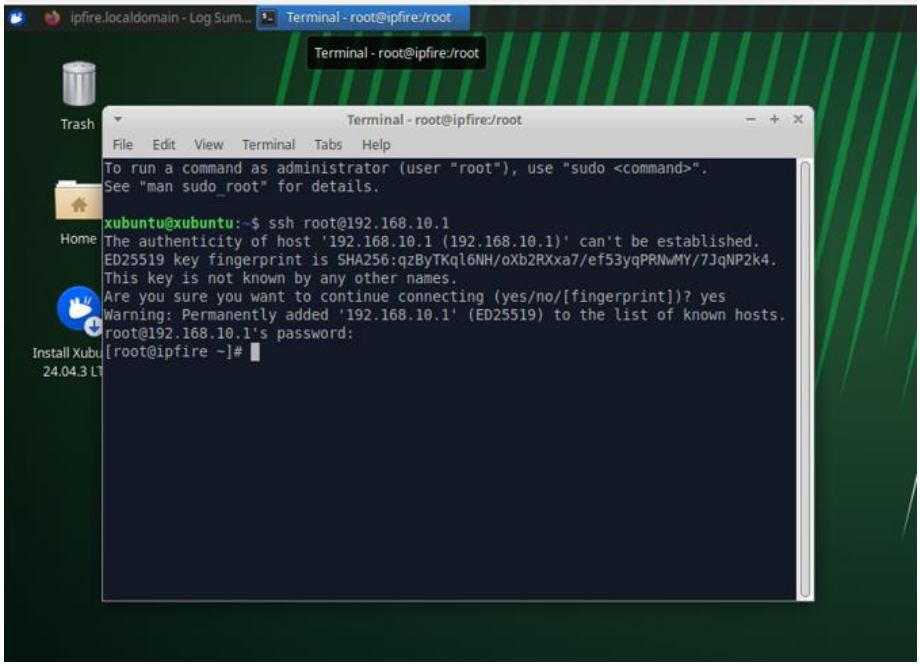
Unlucky for us, SSH wasn't installed on IPFire- more troubleshooting! We went with 'pakfire install openssh' and checked the config file



SSH seemed to be ready...



We tried to restart ssh and see if it would work but it still wasn't being recognized. We ran a simple 'touch /var/ipfire/remote/enablessh' command and finally got in.

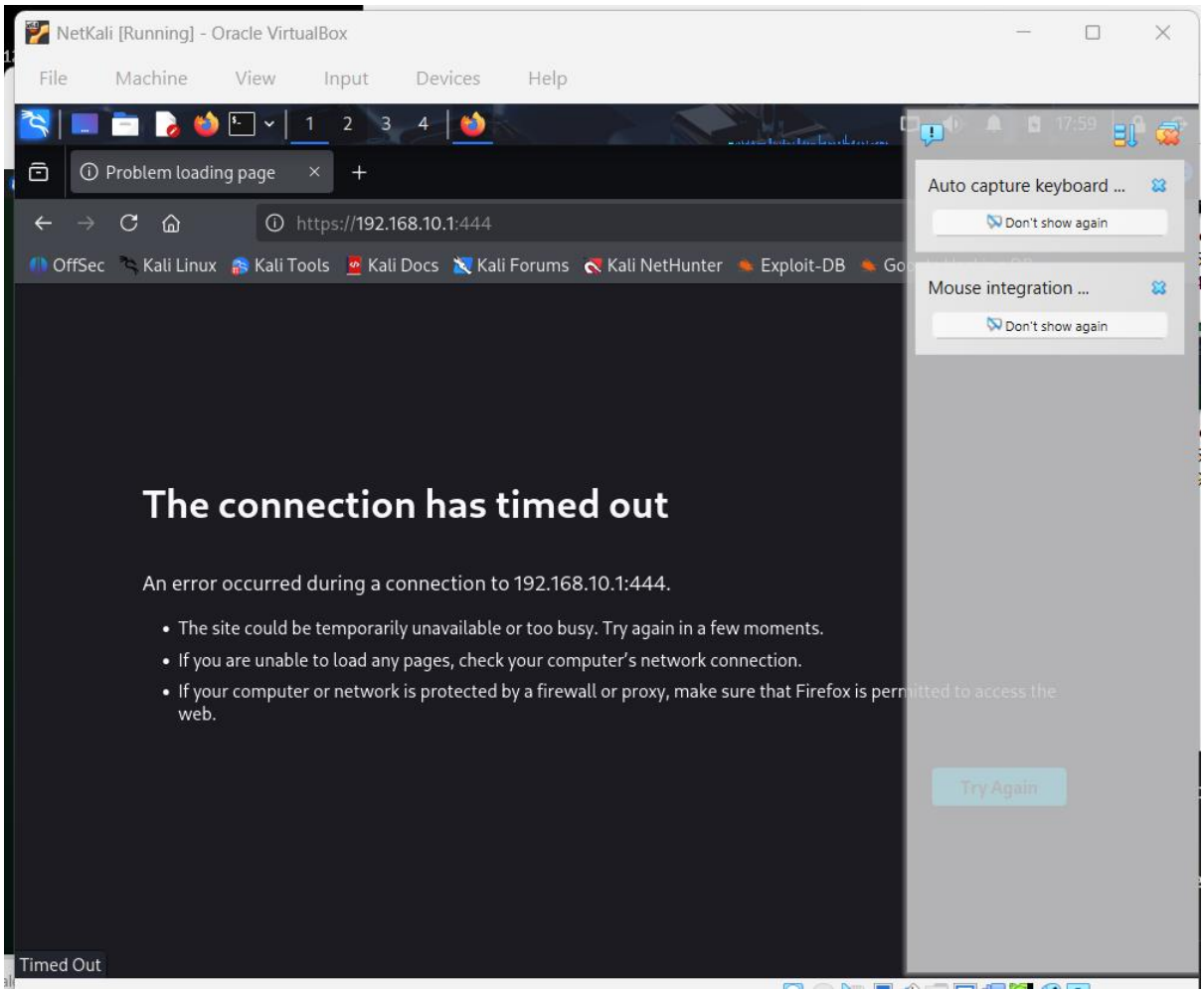


Now to add our security rule to allow the victim but block the attacker.

```
[root@ipfire ~]# iptables -I INPUT 1 -p tcp --dport 444 -s 192.168.10.52 -j ACCE
PT
[root@ipfire ~]# iptables -I 2 -p tcp --dport 444 -s 192.168.10.51 -j DROP
```

```
[root@ipfire ~]# iptables -I INPUT 2 -p tcp --dport 444 -s 192.168.10.51 -j DROP
[root@ipfire ~]# iptables -L INPUT -n --line-numbers
Chain INPUT (policy DROP)
num  target     prot opt source               destination
1    ACCEPT     tcp  --  192.168.10.52        0.0.0.0/0            tcp dpt:444
2    DROP       tcp  --  192.168.10.51        0.0.0.0/0            tcp dpt:444
```

We ensured the rule worked in our logs and double checked by attempting to access.



Now we want to block the attacker from pinging the victim.

```
[root@ipfire ~]# iptables -I INPUT 1 -p icmp --icmp-type echo-request -s 192.168.10.51 -d 192.168.10.52 -j LOG --log-prefix 'KALI_PING: '
[root@ipfire ~]# iptables -I INPUT 2 -p icmp --icmp-type echo-request -s 192.168.10.51 -d 192.168.10.52 -j DROP
```

We ran this command and attempted a ping that resulted in exactly what we wanted.

Before vs After:

```
┌──(sean㉿netkali)-[~]
└─$ ping -c 6 192.168.10.52
PING 192.168.10.52 (192.168.10.52) 56(84) bytes of data.
64 bytes from 192.168.10.52: icmp_seq=1 ttl=64 time=1.28 ms
64 bytes from 192.168.10.52: icmp_seq=2 ttl=64 time=1.50 ms
64 bytes from 192.168.10.52: icmp_seq=3 ttl=64 time=1.49 ms
^[[B^[[B^[[B64 bytes from 192.168.10.52: icmp_seq=4 ttl=64 time=1.04 ms
64 bytes from 192.168.10.52: icmp_seq=5 ttl=64 time=1.02 ms
64 bytes from 192.168.10.52: icmp_seq=6 ttl=64 time=0.959 ms

── 192.168.10.52 ping statistics ──
6 packets transmitted, 6 received, 0% packet loss, time 5014ms
rtt min/avg/max/mdev = 0.959/1.214/1.495/0.220 ms
```

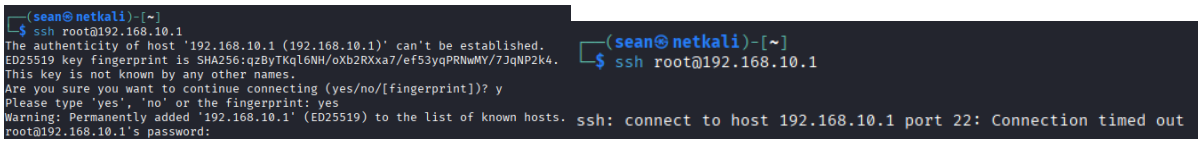
```
┌──(sean㉿netkali)-[~]
└─$ ping -c 6 192.168.10.52
PING 192.168.10.52 (192.168.10.52) 56(84) bytes of data.

── 192.168.10.52 ping statistics ──
6 packets transmitted, 0 received, 100% packet loss, time 5113ms
```

At this point we were running low on time but we had to try one more protective rule, blocking the attacker from the ability to SSH- leaving us with the attacker being unable to reach the Firewalls GUI, the victim with pings and the Firewalls CLI.

```
[root@ipfire ~]# iptables -I INPUT 1 -p tcp --dport 22 -s 192.168.10.51 -j DROP
[root@ipfire ~]# _
```

Once again applying the rule itself was a simple a straighforward process, we attempted to SSH from the attacker.



Here we see a successful result from the before and after.

The rule we added blocked all TCP traffic on port 22 from the attackers IP address. The command forced the firewall to intercept and drop the packets, resulting in a timeout.

To Conclude our experience with IPFire was quite bumpy to start with repeated setbacks due to both personal user error and a bit of unreliability from the firewall itself. Besides these errors it ended up being a valuable lesson in understanding IPFire and how it works. Although the GUI didn't want to work with us and ended up burning time the CLI was a great back up alternative. The project demonstrated how IPFire can block and manage traffic succesfully but with a bit more manual configuration

Conclusion

This project shows the difference between all three open-source firewalls, when tested in a controlled environment. Each firewallhad successfully blocked malicious access when the where given the correct ruleset, showing that open-source firewalls can provide a decent baseline of protection.

We think pfSense was the most sturdy and user-friendly out the bunch, as it offered very detailed logging and a user friendly GUI with very effective rule handeling. OPNsense had similar benefits but had a few lackluster faults, with the rule handeling and overall intrusion prevention as the setup was very tedious and required kind of outdated methods. IPfire in comparison, was a powerful firewall, but it was very unintuitive and the configuration was very tedious to do, most likely due to the use of VM's and the strain it had on the laptop.

Overall, each firewall met the very basic security objectives set, but pfSense stood out amongst the crowd.

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